# Three-way assignment can separate fair core selections maximally

Internal research note

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**Abstract**

We construct standard three-sided assignment games with nonempty entire cores, equally sized sectors, and triple values in $\{0, 1, 2, 3\}$ whose full leximin and full descending-lexicographic-minimum payoffs have normalized $\ell_1$ distance tending to two. This is the sharp supremum. The same construction rules out every positive dimension-independent lower guarantee for a player's leximin payoff relative to their maximum core payoff. A companion family has exponentially vanishing individual guarantees but coincident selectors. The proof uses explicit integral matchings and the entire core, with every fixed-zero player counted. Correctness has been checked internally; novelty and external significance remain subject to specialist review.

## 1 Statement and model

Let $A, B, C$ be finite disjoint sets and $w_{abc} \geq 0$. A coalition's worth is the maximum weight of an *integral* matching of disjoint triples within it. Write $T > 0$ for the grand coalition's matching value. Its entire core is

$$K = \{p \geq 0 : \sum_i p_i = T, \quad p_a + p_b + p_c \geq w_{abc} \text{ for every } (a, b, c)\}.$$

Indeed, the core implies each triple inequality, while summing these inequalities over a coalition's optimal matching proves every coalition inequality. We assume $K \neq \varnothing$; nonemptiness does not hold for arbitrary arrays. Let $x$ maximize the ascending sorted payoff vector lexicographically, and let $z$ minimize the descending sorted vector lexicographically. Both are full selectors; fixed zeros remain in the sorted vectors. Set $\rho = \|x - z\|_1 / T$.

**Theorem 1.** *For every integer $L \geq 1$ and $R \geq 0$, there is such a game with*

$$n = 9(L + R) + 12, \qquad T = 3(L + R + 2), \qquad w_{abc} \in \{0, 1, 2, 3\},$$

*and*

$$\rho(L, R) = \frac{2((R + 2)2^L - 1)}{(2^L + 2)(L + R + 2)}. \tag{1}$$

*Consequently the sharp supremum of $\rho$ over three-sided assignment games with nonempty entire cores is 2. A player in this family has maximum core payoff 1 but leximin payoff $2/(2^L + 2)$, so no positive dimension-independent individual fraction survives.*

The two-sided predecessor establishes $x_i \geq y_i/2$ for every core point $y$ and sharp maximal distance $\lfloor n/2 \rfloor / (\lfloor n/2 \rfloor + 1)$ for $n \geq 4$; that internal result is preserved separately. Theorem 1 places the change at team size three, even with bounded integer values. It neither asserts a sharp finite-$n$ formula nor resolves the best convergence rate.

## 2 An explicit market

All color indices are modulo three. Introduce global players $G_i$ of color $i$. At levels $\ell = 0, \dots, L$ introduce $U_{\ell i}, V_{\ell i}, D_{\ell i}$ for $i = 0, 1, 2$. At even levels their colors are $i, i+1, i+2$; at odd levels the colors of $U$ and $V$ are interchanged. Put $s_0 = 2$ and $s_\ell = 1$ for $\ell \geq 1$. Assign the following positive triple values:

| Triple | Value |
|---|---|
| $\{U_{\ell i}, V_{\ell i}, D_{\ell i}\}$ | $s_\ell$ |
| $\{U_{\ell i}, V_{\ell i}, G_{i+2}\}$ | $s_\ell$ |
| $\{U_{0i}, G_{i+1}, G_{i+2}\}$ | 1 |
| $\{U_{\ell i}, V_{\ell-1,j}, V_{\ell-1,k}\}$, $\ell \geq 1$ | 2 |
| $\{V_{\ell i}, U_{\ell-1,j}, U_{\ell-1,k}\}$, $\ell \geq 1$ | 3 if $\ell = 1$, else 1 |

In the last two rows, $\{j, k\} = \{0, 1, 2\} \setminus \{i\}$. Each listed triple has exactly one player of each color.

For every $i$ introduce $R$ copies $(U'_{ri}, V'_{ri}, D'_{ri})$ with the same colors as its top-level triple. Give value 1 to each copy's local and global pair triples $\{U'_{ri}, V'_{ri}, D'_{ri}\}$ and $\{U'_{ri}, V'_{ri}, G_{i+2}\}$, and to its cross-triples

$$\{U'_{ri}, V_{Li}, G_{i+2}\}, \qquad \{V'_{ri}, U_{Li}, G_{i+2}\}.$$

Every unlisted triple has value zero. There are $3(L + R) + 4$ players per sector.

## 3 Integral value and entire core

The triples using each pair's own $D$ form a disjoint integral matching with weight $6 + 3L + 3R = T$. Consider the allocation assigning 1 to both root $U$ and root $V$, 0 to every subsequent $U$, 1 to every subsequent $V$, the analogous values to copies, and 0 to every dummy. It has total $T$ and covers every weighted triple. Summing its payoffs over any disjoint matching bounds that matching's weight by $T$. Hence the displayed integral matching is optimal and this allocation belongs to the entire core.

**Lemma 1.** *Every core allocation, and only a core allocation, has all dummy payoffs zero and the following form:*

$$\begin{aligned} U_{0i} &= 1 + q_{0i}, & V_{0i} &= 1 - q_{0i}, \\ U_{\ell i} &= q_{\ell i}, & V_{\ell i} &= 1 - q_{\ell i} \quad (\ell \geq 1), \\ U'_{ri} &= q_{Li}, & V'_{ri} &= 1 - q_{Li}, \end{aligned}$$

*where* $0 \leq q_{\ell i} \leq 1$ *and*

$$q_{\ell i} = q_{\ell-1,j} + q_{\ell-1,k}, \qquad q_\ell = (J - I) q_{\ell-1}. \tag{2}$$

*Here* $J$ *is the* $3 \times 3$ *all-ones matrix.*

*Proof.* Sum the inequalities of the displayed matching. Efficiency and nonnegativity force every unmatched $G$ to zero and every matching triple to equality. The alternative pair triple using $G$ then forces its local $D$ to zero and its pair to total $s_\ell$. Root lower-bound triples give $U_{0i} \geq 1$. At level one, the two recurrence inequalities become respectively $q_{1i} \geq q_{0j} + q_{0k}$ and $q_{1i} \leq q_{0j} + q_{0k}$ after substituting the root offsets. At subsequent levels the same substitutions give these opposite inequalities without offsets. Cross-triples give opposite inequalities between a copied $U$ and its original top $U$, hence equality. Nonnegativity supplies the stated bounds.

Conversely, the parameterization satisfies every displayed triple inequality, has total $T$, and is nonnegative. All omitted zero-weight triple inequalities are automatic. It therefore satisfies every coalition inequality by summing over a matching. This proves equality with the entire core. □

## 4 The full selectors and quantitative separation

For minimax, every root $U$ is at least 1. The allocation just used, denoted $z$, has maximum payoff 1. Any core allocation with maximum at most 1 must have $U_{0i} = 1$ for all $i$, hence $q_0 = 0$. Lemma 1 then fixes every coordinate. Thus $z$ is the unique minimax allocation and the full descending-lexicographic minimum.

For leximin, all $D, G$ are fixed at zero. Every other coordinate can simultaneously be positive by taking $q_0 = t(1,1,1)$ with $0 < t < 2^{-L}$. Consequently the full procedure first fixes exactly the dummy coordinates at zero, then maximizes a common floor $\eta$ for all remaining coordinates. If such a floor is feasible, level-one $U$ and top-level $V$ imply

$$2\sum_i q_{0i} = \sum_i q_{1i} \geq 3\eta, \qquad 2^L \sum_i q_{0i} = \sum_i q_{Li} \leq 3(1-\eta).$$

Combining gives $\eta \leq 2/(2^L+2)$. Equality is attained by

$$t = \frac{1}{2^L+2}, \qquad q_{\ell i} = 2^\ell t. \tag{3}$$

The least nonroot $U$ is $2t$ and the least $V$ is $1 - 2^L t = 2t$; root and copy payoffs also respect this floor. Equality forces $q_{1i} = \eta$ for all three $i$. Since $J - I$ is invertible, with eigenvalues $2, -1, -1$, it then forces $q_0 = t(1,1,1)$. Thus the optimal positive-floor face is a singleton and (3) gives the full leximin allocation $x$. No symmetry restriction on the feasible core was imposed.

Every level-$\ell$ pair changes from $z$ to $x$ by $(2^\ell t, -2^\ell t)$, including the offset root pair. Each copied pair changes by $(2^L t, -2^L t)$. Therefore

$$\|x - z\|_1 = 6t\left(\sum_{\ell=0}^{L} 2^\ell + R2^L\right) = \frac{6((R+2)2^L - 1)}{2^L+2}.$$

Dividing by $T$ proves (1). With $R = L^2$, both the amplified payoff fraction and the share of copied pairs tend to one, giving $\rho \to 2$. Conversely, nonnegative vectors of total $T$ always satisfy $\|x - z\|_1 \leq 2T$, proving the sharp supremum. For example, $L = 4, R = 16$ gives $\rho = 287/198 > 4/3$. Finally, every top $V$ has core maximum 1, attained at $z$, but payoff $2/(2^L+2)$ at $x$. This proves the individual conclusion separately.

**Proposition 1.** *Let $S_n$ be the supremum for total player count $n$, allowing arbitrary sector sizes. For $n \geq 30$, put $k = \lfloor (n-12)/9 \rfloor$. Then*

$$0 \leq 2 - S_n \leq \frac{2\lceil \log_2 k \rceil + 4}{k+2} + \frac{2}{(k+2)^2} = O(\log n / n).$$

*Proof.* Choose $L = \lceil \log_2 k \rceil$ and $R = k - L \geq 0$, and add the remaining at most eight players with zero incident values. Matching efficiency forces these players to zero in the entire core, preserving both full selectors. Subtracting (1) from two gives

$$2 - \rho = \frac{2L2^L}{(2^L+2)(k+2)} + \frac{4}{2^L+2} + \frac{2}{(2^L+2)(k+2)}.$$

Use $2^L \geq k$ and bound the first fraction by $2L/(k+2)$. □

## 5 Controls, calibration, and scope

The construction separates two mechanisms. Recurrence amplifies payoff differences across levels. Root pair offsets pin the minimax allocation; top copies make the amplified changes dominate total value. Taking $R = 0$ and $L \to \infty$ makes both the individual ratio and $\rho$ tend to zero. Taking fixed $L$ and $R \to \infty$ gives only $\rho \to 2^{L+1}/(2^L + 2) < 2$.

There is also an exact coincident-selector control. Set every pair total to 1, remove root lower-bound triples and root offsets, and change the level-one complementary triple value from 3 to 1. With no copies the same proof gives $U_\ell = (J - I)^\ell U_0$, $V_\ell = 1 - U_\ell$. A positive floor $\eta$ gives $\sum_i U_{0i} \geq 3\eta$ and $2^L \sum_i U_{0i} \leq 3(1 - \eta)$, hence $\eta \leq 1/(2^L + 1)$. A ceiling $M$ gives $\sum_i U_{0i} \geq 3(1 - M)$ and $2^L \sum_i U_{0i} \leq 3M$, hence $M \geq 2^L/(2^L + 1)$. Both bounds are attained by $U_{\ell i} = 2^\ell/(2^L + 1)$; equality in each pins all root coordinates and therefore the entire allocation. Thus $x = z$, while a top $V$ has selected payoff $1/(2^L + 1)$ and core maximum 1. The proposed one-third individual guarantee fails already at $L = 2$, with 30 players.

The additive calibration $w_{ijk} = 1 + \mathbf{1}_{i=1} + \mathbf{1}_{j=1}$ with $m \geq 2$ players per sector has $T = m + 2$ and core sector shifts $(t_A, t_B, t_C) \geq 0$, $t_A + t_B + t_C = 1$, on top of the two exceptional unit payoffs. All triples lie in optimal perfect matchings and are tight, which proves this entire-core description. Leximin uses shifts $(1/3, 1/3, 1/3)$ and minimax $(0, 0, 1)$, giving $4m/[3(m + 2)]$. This benchmark is exceeded by Theorem 1.

Exact rational checks cover all 256 binary $2 \times 2 \times 2$ arrays, the sourced empty-core example below, twelve deterministic nonadditive $3 \times 3 \times 3$ arrays, additive cases, and raw recurrence families through 192 players. Integral primal matchings and feasible covering allocations certify the larger examples' matching totals before full staged selector optimization. Separate implementations check the construction and the unshifted control. These finite checks support the conventional proof; they are not the asymptotic argument.

For literature calibration, the published Atay–Núñez example has layers $\left(\begin{smallmatrix} 3 & 1 \\ 2 & 5 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 4 \\ 5 & 4 \end{smallmatrix}\right)$, with unique diagonal matching total 7. Lucas's two conditions, as reproduced there, are $2w_{111} + w_{222} \geq w_{112} + w_{121} + w_{211}$ and $w_{111} + 2w_{222} \geq w_{221} + w_{212} + w_{122}$, conditional on an optimal diagonal. Here the second reads $11 \geq 14$, so the core is empty. The exact calibration reproduces this failure.

Quint initiated the multisided core model; assortative and graph-based subclasses impose additional restrictions and do not supply a bound for arbitrary triple arrays. The dated companion source audit found no matching prior theorem in its bounded search. Full original Quint and Lucas articles were not available in that audit. Accordingly, this note records an internally proved structural result, not a certified novelty or field-level breakthrough.

## References


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